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PDMark: Plausibly deniable watermarking for diffusion models

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ABSTRACT

Diffusion watermarking has become an important technique for copyright protection and provenance tracing in AI-generated content. However, most existing methods rely on deterministic extraction, where the embedded watermark is expected to be revealed once the corresponding key or extractor is available. This design becomes problematic under coercive disclosure, where revealing a watermark may expose sensitive identity or provenance information.

In this paper, we propose PDMark, a plausibly deniable watermarking method for diffusion models. To the best of our knowledge, PDMark is the first framework that explicitly studies deniable watermarking for diffusion-generated images under coercive disclosure. PDMark is a train-free latent-based framework, and it constructs a single watermark representation from the real and fake messages. The representation is then mapped to a watermarked latent through Gaussian-consistent sampling. During extraction, the disclosed key determines which watermark is recovered, allowing the content owner to disclose a fake extraction key that produces a plausible fake watermark while the undisclosed watermark remains protected. Extensive experiments and theoretical analysis demonstrate that PDMark realizes deniability while maintaining high extraction accuracy, competitive image fidelity, and robustness under various attacks.

1. Introduction

As diffusion models are increasingly adopted for image generation and editing, AI-generated content has become prevalent in creative media production (Chen et al., 2024). The outstanding generation quality and flexibility have made diffusion models effective in tasks such as text-to-image generation (Rombach et al., 2022), image editing (Brooks et al., 2023) and restoration (Saharia et al., 2023). As AI-generated content grows in real-world scenarios, copyright protection and traceability without compromising normal usability have become critical requirements. Generated images may require watermarks to support ownership verification and copyright claims (Fernandez et al., 2023; Lei et al., 2025; Tian & Wang, 2026), or to enable responsibility tracing when generated content is used for deepfakes or fabricated events (Liu et al., 2023a; Siddiqui et al., 2025; Wu et al., 2021).

Diffusion watermarking has attracted widespread research attention (Kong et al., 2026; Liu et al., 2026), and recent relevant methods mainly include model-tuning-based ones and latent-based ones (Wan et al., 2022). Model-tuning-based methods modify the diffusion model itself (Fernandez et al., 2023; Min et al., 2024; Wang et al., 2025b), while latent-based methods embed watermark into its latent domain (Wen et al., 2023; Yang et al., 2024). Despite their differences in design, these

methods mainly focus on the objectives of model fidelity, extraction accuracy, and robustness (Wang et al., 2026). However, existing methods are designed for deterministic verification, and given a watermarked image, the extraction process is expected to reveal the embedded watermark.

In other words, existing methods assume that watermark verification is always benign, but such an assumption may fail when the watermark carries sensitive identity or provenance information (Min et al., 2024; Xu et al., 2022). In some scenarios, the watermark may carry sensitive information, such as the identity of a protected content creator, a confidential user identifier, or the transmission path of a sensitive file. Once an adversary obtains the watermarked image and coerces the owner to reveal the watermark, ordinary watermarking methods can only disclose the real watermark. Therefore, the security goal extends beyond robust extraction, and it requires a practical watermarking system should also prevent sensitive provenance information from being exposed under coercive disclosure. It is worth noting that even multi-watermark methods, such as RingID (Ci et al., 2024), offer limited protection in this setting. Their goal is to distinguish different users associated with an image, rather than to hide whether another meaningful watermark exists after one extraction key has been disclosed. Therefore, multi-key identification and deniable disclosure are fundamentally different objectives. As

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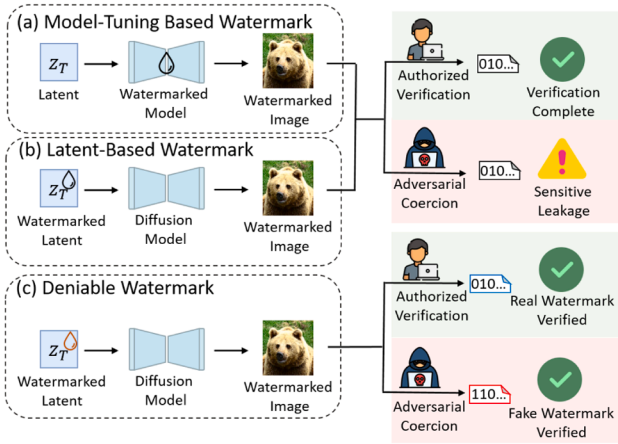


Fig. 1. Comparison of watermarking methods under adversarial coercion.

a result, conventional diffusion watermarking lacks a mechanism for protecting sensitive watermark content once coercive disclosure occurs.

Deniability offers a promising solution to address this limitation. For example, an enterprise may protect a confidential client-specific identifier while disclosing a general organizational provenance watermark. Similarly, a sensitive document-distribution system may conceal a route-specific tracing identifier by disclosing only a non-sensitive source-verification watermark. This mechanism reduces disclosure risk under the specified threat model but does not replace legal protection or secure key management. Originally introduced in cryptography by [Canetti et al. \(1997\)](#), deniability allows a coerced user to disclose a fake message from the observed carrier as a response to adversarial coercion, thereby protecting both the true secret and the coerced user. Deniability has also been studied in image hiding, where a user under coercion may reveal a fake secret while keeping the sensitive secret concealed ([Xu & Xia, 2023](#); [Xu et al., 2022](#)). [Liu et al. \(2023b\)](#) introduced deniable generative steganography for diffusion-based information hiding, which permits the same image to be decoded with a fake message while keeping the existence of the real message concealed. However, its deniability relies on the selection of two different decoders, making the decoder itself part of the secret component, and maintaining multiple decoders introduces extra training costs and disclosure risks, limiting its applicability to real-world deployments. [Fig. 1](#) illustrates the comparison of existing diffusion watermark methods and their performance under coercion scenarios.

Addressing coercive disclosure therefore calls for a new form of diffusion watermarking that incorporates deniability as a core security requirement. Unlike conventional diffusion watermarking, which mainly aims at reliable ownership verification or provenance tracing, deniable watermarking must also protect sensitive watermark information when the owner is coerced to disclose an extraction key. This introduces a new security requirement that the same watermarked image should support key-dependent extraction results, while the disclosed key should not expose whether another meaningful watermark exists. A naive solution that simply embeds two ordinary watermarks into the same image cannot satisfy this requirement, because the undisclosed watermark may remain as an additional structured signal, making it easier to determine the existence of another watermark after one watermark is disclosed. The challenge is further amplified in latent diffusion watermarking, where generation starts from Gaussian noise and deviations from the standard Gaussian latent distribution may degrade image quality or expose watermark traces ([Ci et al., 2024](#); [Yang et al., 2024](#)). Therefore, deniable diffusion watermarking requires a fundamentally different design that can provide different extraction results under different keys without compromising the statistical behavior and practical reliability of the watermark.

To address this limitation, we propose PDMark, a latent-based plausibly deniable watermarking method for diffusion models. To the best of our knowledge, PDMark is among the first frameworks to explicitly introduce plausible deniability into diffusion model watermarking. PDMark allows a single watermarked image to reveal one of two distinct watermarks under different extraction keys, while concealing the existence of the undisclosed watermark. In normal case, the real key is used to extract the real watermark, while under coercion, the fake key can be disclosed to extract a plausible fake watermark. Our method train-freely constructs a single watermark representation for both messages with a synchronization header to stabilize recovery under distortion. The watermarked latent is sampled according to the watermark representation through a Gaussian-consistent sampling strategy. During extraction, the disclosed key reveals only its corresponding watermark without revealing useful information about the undisclosed watermark.

Our contributions are summarized as follows:

- We propose a deniable watermarking method for diffusion models, in which the same watermarked image supports deniable extraction results under different keys, while an invalid key reveals no meaningful result.
- We develop a train-free latent watermarking technique that constructs a single watermark representation for the real and fake messages, which combines with synchronization-assisted extraction, and Gaussian-consistent latent sampling.
- Extensive experiments demonstrate strong extraction accuracy, fidelity, robustness, and deniability.

2. Related work

2.1. Diffusion model watermarking

Recent watermarking methods for diffusion models can be broadly categorized into model-tuning-based ones and latent-based ones. Model-tuning-based methods embed watermark in the diffusion process by fine-tuning model components, so that naturally generate images with watermark. Stable Signature ([Fernandez et al., 2023](#)) embeds a binary watermark in the VAE decoder of a latent diffusion model. Flexible and Secure Watermarking for Latent Diffusion Model ([Xiong et al., 2023](#)) studies flexible message-specific watermark embedding for latent diffusion models. ProMark ([Asnani et al., 2024](#)) emphasizes causal attribution of the watermark. WaDiff ([Min et al., 2024](#)) conditions the diffusion process on user-specific watermark for content attribution. WMAAdapter ([Ci et al., 2025](#)) introduces a plug-and-play watermark control mechanism for latent diffusion models. SleeperMark ([Wang et al., 2025b](#)) studies backdoor watermark robustness under downstream fine-tuning of text-to-image diffusion models.

Latent-based watermarking methods embed watermark into the initial or in-process latent noise, without substantially fine-tuning the model. Tree-Ring Watermarks ([Wen et al., 2023](#)) popularize inversion-based watermarking by encoding structured patterns in the initial noise and recovering them through diffusion inversion. Gaussian Shading ([Yang et al., 2024](#)) proposes a training-free and distribution-preserving watermarking scheme with theoretical analysis of performance losslessness. RingID ([Ci et al., 2024](#)) revisits Tree-Ring watermarking for enhanced multi-key identification. LaWa ([Rezaei et al., 2024](#)) performs in-generation watermarking by modifying latent-space decoding in latent diffusion models. Meng et al. (2025) injects and detects watermark directly in latent diffusion space, and uses a progressive training strategy to alleviate the trade-off between image quality and robustness. MarkPluggger ([Zhang et al., 2025](#)) proposes a plug-and-play latent watermarking framework without retraining, and improves transferability by fusing watermark representations that are approximately orthogonal to image semantics across different LDM variants. CoSDA ([Fang et al., 2025](#)) improves the robustness of inversion-based generative watermarking while maintaining visual quality, and is de-

signed as an enhancement to the extraction side of latent watermark recovery. GaussMarker (Li et al., 2025) introduces a dual-domain watermarking strategy that embeds watermark in both the spatial and frequency domains, together with a Gaussian Noise Restorer to improve robustness against distortions and stronger attacks. ROAR (Wang et al., 2025a) reduces inversion errors in latent-based watermarking through regeneration-based optimization and distortion-adaptive restoration, thereby strengthening extraction robustness under both latent and pixel domain perturbations. InvisMark (Xu et al., 2025a) emphasizes invisible yet high-capacity provenance marking for AI-generated images, achieving robust decoding of long payloads such as 256-bit UUID-style identifiers under common image manipulations. TAG-WM (Chen et al., 2025) extends latent watermarking to tamper-aware settings by jointly embedding copyright and localization watermarks, so that robust decoding and tamper localization can be performed together through diffusion inversion sensitivity. Recent studies show that diffusion watermarks may be forged, removed, or manipulated under realistic black-box or adaptive threat models (Müller et al., 2025; Yuan et al., 2025).

2.2. Deniability in image hiding

Deniability was first studied in cryptography through deniable encryption, where a party under coercion can provide an alternative explanation for an observed ciphertext (Canetti et al., 1997). This idea later inspired image-domain hiding under coercive disclosure, where the goal is not only to conceal a secret, but also to make an alternative secret appear plausible.

Early work such as Digital-Invisible-Ink (Huang et al., 2008) already explored the possibility of plausible alternative embedded content. More recent studies made the deniability objective more explicit. Receiver-deniable steganography (Xu et al., 2022) formalized the setting in which a receiver can reveal a fake secret under coercion, while sender-deniable image steganography (Xu & Xia, 2023) studied how a sender can reproduce the same stego image using a fake secret or a fake key. Ahmad and Rass (2023) further analyzed when such fake openings remain statistically and semantically convincing under adversarial inspection. At the system level, SMPDE (Xia et al., 2024) showed that plausible deniability is also practically relevant when image hiding is combined with steganography, watermarking, and higher-layer secure mechanisms.

Recent generative methods extend covert communication to diffusion-based image synthesis (Zhou et al., 2025). Deniable Diffusion Generative Steganography (DDGS) (Liu et al., 2023b) is the most closely related work in this direction. It introduces deniability into diffusion-based generative hiding by embedding both real and fake messages and realizing the two extractions through separate decoders. More generally, diffusion-based generative steganography has rapidly evolved from invertible or carrier-free generation schemes (Kim et al., 2023; Wei et al., 2023) to stronger diffusion hiding pipelines with improved visual fidelity and extraction reliability.

3. Preliminary

In this section, we briefly review the formulation of Denoising Diffusion Implicit Models and state the threat model.

3.1. Denoising diffusion implicit models

Diffusion models generate images by progressively transforming Gaussian noise into structured visual content. Let $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ denote a clean sample, and let $\mathbf{x}_{1:T}$ denote its noisy versions over T timesteps. Given a variance schedule $\{\beta_t\}_{t=1}^T$, define $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. Then, the marginal distribution of the forward diffusion process can be written as

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (1)$$

DDIM defines a non-Markovian reverse process that enables deterministic sampling while preserving the standard noise-prediction training objective. Given a noisy sample $\mathbf{x}_t$, the network $\epsilon_\theta(\mathbf{x}_t, t, \mathbf{c})$ predicts the noise, following Eq. (2).

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t, \mathbf{c})}{\sqrt{\bar{\alpha}_t}}. \quad (2)$$

The corresponding DDIM transition is

$$\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \epsilon_\theta(\mathbf{x}_t, t, \mathbf{c}) + \sigma_t \mathbf{z}, \quad (3)$$

where $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. In the deterministic case, $\sigma_t = 0$. Starting from $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, repeated application of this update produces the final sample $\mathbf{x}_0$.

The model is trained with the standard noise-prediction objective

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\mathbf{x}_0, t, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t, \mathbf{c})\|_2^2]. \quad (4)$$

In practical text-to-image systems, diffusion is usually performed in the latent space rather than directly in pixel space, where the latent code is given by $\mathbf{z}_0 = \mathcal{E}(\mathbf{x})$. The forward noising process becomes

$$\mathbf{z}_t = \sqrt{\bar{\alpha}_t} \mathbf{z}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (5)$$

while the reverse DDIM process progressively reconstructs $\hat{\mathbf{z}}_0$, which is then decoded back to the image space by $\hat{\mathbf{x}} = \mathcal{D}(\hat{\mathbf{z}}_0)$. Since our method operates on the latent variables used by the diffusion process, this latent DDIM formulation serves as the basis for watermark embedding and extraction.

3.2. Threat model

We consider deniable diffusion watermarking under coercive disclosure, where the content owner may be forced by the adversary to reveal the watermark associated with an obtained image.

- Let Alice denote the content owner, who embeds watermark into the generated image and may further be coerced for watermark revealing.
- Let Bob denote an authorized auditor, who verifies the watermark using the real extraction key.
- Let Carol denote an unauthorized adversary, who obtains the watermarked image and attempts to acquire watermark information by randomly guessing or coercing Alice.

In this setting, deniability is defined against Carol, while Bob represents the benign verification party.

Carol is assumed to know the conventional diffusion watermarking workflow and all public parameters. Thus, once Alice discloses one extraction key, Carol can run the extraction procedure with the disclosed key and verify the corresponding watermark. Carol may also exploit the public information to guess extraction key and apply the extraction procedure to acquire watermark information. Before any extraction key is disclosed, Carol should not be able to identify the watermark or reliably distinguish a watermarked image from an ordinary generated image.

We consider coercion after observation of the watermarked image, where Carol first obtains the image and may then coerce Alice to reveal one extraction key for that image, in analogy to the post-communication coercion setting in deniable encryption (Canetti et al., 1997). In normal verification, Alice or Bob uses the real extraction key $k^{(r)}$ to extract the real watermark $\mathbf{m}^{(r)}$. Under coercion, Alice may disclose the fake extraction key $k^{(f)}$, and Carol extracts a semantically consistent and verifiable fake watermark $\mathbf{m}^{(f)}$ with the disclosed key. Carol may also apply the extraction procedure under arbitrary guessed keys, including keys that reuse any disclosed common component but replace the private component with a guessed one.

We focus on prepared deniability under single-key coercive disclosure (Chakraborti et al., 2017). The real and fake watermarks are prepared during watermark construction, and Alice is coerced to reveal

one extraction key for the obtained image rather than all secret materials associated with the image. Because Carol can verify the watermark extracted with the disclosed key, the disclosure effectively provides a valid key–watermark pair and naturally induces a known-plaintext attack setting.

Under this post-disclosure setting, Carol has the watermarked image, the public nonce, all public algorithms and hyperparameters, the disclosed key $k^{(r)} = (k_c, k_r)$, the corresponding watermark plaintext, and every quantity derivable from them. Using this information, Carol attempts to distinguish a meaningful undisclosed branch from a same-form auxiliary branch, distinguish a dual-branch carrier from a single-branch carrier, or reconstruct the undisclosed ciphertext or plaintext bits, without access to k_r . Image-level side channels and external metadata not represented in the carrier remain outside this claim and are discussed as limitations.

Our security objective has four aspects.

- **Pre-disclosure undetectability:** Without disclosed key, Carol should not identify the watermark or reliably distinguish watermarked images from ordinary generated images.
- **Content confidentiality:** Disclosing one extraction key should not reveal the content of the undisclosed watermark.
- **Disclosure ambiguity:** Given one extracted watermark, Carol should not determine whether it corresponds to the real or fake one.
- **Post-disclosure existence ambiguity:** After one key is disclosed, Carol should not distinguish whether the undisclosed watermark carries a meaningful information or auxiliary random padding.

4. Method

4.1. Overview

The workflow of PDMark is shown in Fig. 2. Our method is trainable, and a single deniable watermark representation is constructed before embedding. Given a pair of real and fake messages $(\mathbf{m}^{(r)}, \mathbf{m}^{(f)})$ and the corresponding keys $(k^{(r)}, k^{(f)})$, we first construct a single watermark representation, which combines a key-derived synchronization component with a data component. And then we sample a watermarked latent $\tilde{\mathbf{z}}$ from it for diffusion generation:

$$\tilde{\mathbf{z}} = \mathcal{N}(\mathbf{m}^{(r)}, \mathbf{m}^{(f)}, k^{(r)}, k^{(f)}), \quad (6)$$

and the final image $\tilde{\mathbf{x}}$ is generated from $\tilde{\mathbf{z}}$ with DDIM.

The watermark is extracted from the obtained image $\tilde{\mathbf{x}}$ with the corresponding key after correction by the synchronization component:

$$\hat{\mathbf{m}}^{(r)} = \mathcal{R}(\tilde{\mathbf{x}}, k^{(r)}), \quad \hat{\mathbf{m}}^{(f)} = \mathcal{R}(\tilde{\mathbf{x}}, k^{(f)}). \quad (7)$$

PDMark is particularly designed for deniability, and both extractable results are constructed at embedding time. Since the recovered object is a single representation, revealing one extraction does not directly expose whether the undisclosed branch carries another meaningful watermark or only auxiliary padding. In addition, inspired by MDDM (Xu et al., 2025b), we design a Gaussian-consistent latent sampling strategy, and further adapt and optimize it for the watermark setting.

4.2. Watermark construction

Parameters Initialization. We generate the real key $k^{(r)}$ and the fake key $k^{(f)}$. Each key consists of a common component and a private component:

$$k^{(j)} = (k_c, k_j), \quad j \in \{r, f\}, \quad (8)$$

where k_c is shared by both keys, and k_j is the private component of watermark j . Then each private component k_j derives a specific binary relation by the random generator $\mathcal{G}_H$

$$(\mathbf{H}_v^{(j)}, \mathbf{d}_v^{(j)}) = \mathcal{G}_H(k_j, v), \quad j \in \{r, f\}, \quad (9)$$

where $\mathbf{H}_v^{(j)} \in \{0, 1\}^{L \times M_{\text{data}}}$, $\mathbf{d}_v^{(j)} \in \{0, 1\}^L$ and $v \in \{0, 1\}^{l_n}$ is a public nonce.

For every image, we first sample a fresh initial public nonce v_0 . Let $\mathbf{A}_v = ((\mathbf{H}_v^{(r)})^\top, (\mathbf{H}_v^{(f)})^\top)^\top$, which requires $M_{\text{data}} \geq 2L$. PDMark requires $\text{rank}(\mathbf{A}_v) = 2L$ so that every pair of branch targets is solvable. If the condition fails, the nonce is deterministically refreshed from v_0 , both branch relations are regenerated, and construction aborts if the configured refresh limit is exhausted.

Representation Construction. We construct a binary watermark representation $\mathbf{b} \in \{0, 1\}^M$ and partition it into a synchronization header of length L_{sync} and a data carrier of length M_{data} , with $M = L_{\text{sync}} + M_{\text{data}}$.

The synchronization header is used to compensate for distribution drift and geometric distortions before extraction, while the data carrier is embedded with the two watermarks.

The synchronization header is generated by a pseudorandom generator keyed by the common component:

$$\mathbf{h}_v^{\text{sync}} = \text{PRNG}(k_c, v) \in \{0, 1\}^{L_{\text{sync}}}. \quad (10)$$

For each watermark $j \in \{r, f\}$, the message is masked by a nonce-based pseudorandom stream derived from its private component:

$$\tilde{\mathbf{c}}^{(j)} = \mathbf{m}^{(j)} \oplus \mathcal{G}_{\text{enc}}(k_j, v; \text{msg-enc}) \in \{0, 1\}^L. \quad (11)$$

Here, $\mathcal{G}_{\text{enc}}$ is instantiated by a standard nonce-based stream-cipher construction, and msg-enc is a fixed public domain-separation label for message masking. The fresh-nonce rule above prevents stream reuse under the same private key.

We then construct the data carrier $\mathbf{b}^{\text{data}} \in \{0, 1\}^{M_{\text{data}}}$ that simultaneously satisfies the relations:

$$\mathbf{H}_v^{(j)} \mathbf{b}^{\text{data}} = \tilde{\mathbf{c}}^{(j)} \oplus \mathbf{d}_v^{(j)}, \quad j \in \{r, f\}. \quad (12)$$

Let the stacked system and target be

$$\mathbf{A}_v = \begin{bmatrix} \mathbf{H}_v^{(r)} \\ \mathbf{H}_v^{(f)} \end{bmatrix}, \quad \mathbf{y}_v = \begin{bmatrix} \tilde{\mathbf{c}}^{(r)} \oplus \mathbf{d}_v^{(r)} \\ \tilde{\mathbf{c}}^{(f)} \oplus \mathbf{d}_v^{(f)} \end{bmatrix}. \quad (13)$$

Because $\text{rank}(\mathbf{A}_v) = 2L$, the nullity is $M_{\text{data}} - 2L$. Let $\mathbf{b}_0$ be a particular solution and let $\mathbf{N}_v$ contain a basis of the null space of $\mathbf{A}_v$; PDMark samples

$$\mathbf{b}^{\text{data}} = \mathbf{b}_0 \oplus \mathbf{N}_v \mathbf{u}, \quad \mathbf{u} \leftarrow \{0, 1\}^{M_{\text{data}} - 2L}. \quad (14)$$

Equivalently, row reduction samples every free variable independently from Bernoulli(1/2) and solves the pivot variables, making $\mathbf{b}^{\text{data}}$ uniform over the affine solution space. The real and fake watermarks are therefore encoded as two same-form keyed relations over the shared data component.

Finally, we concatenate the synchronization header and the data carrier to form the watermark representation:

$$\mathbf{b} = [\mathbf{h}_v^{\text{sync}}; \mathbf{b}^{\text{data}}] \in \{0, 1\}^M. \quad (15)$$

4.3. Image generation

Given the watermark representation $\mathbf{b} \in \{0, 1\}^M$ and the latent of size (C, H, W) , we sample the watermarked latent $\tilde{\mathbf{z}}$ for image generation.

We generate a coordinate order from the common component and the public nonce:

$$\Omega_v = \mathcal{G}_\Omega(k_c, v; d) = (\omega_1, \dots, \omega_d), \quad (16)$$

where Ω_v is a pseudorandom permutation of the $d = C \times H \times W$ flattened latent coordinates.

We utilize the sign of latent coordinates to embed the watermark representation. Let $\Phi(\cdot)$ and $\phi(\cdot)$ denote the cumulative distribution function and density function of the standard Gaussian distribution, and let k be the truncation threshold. Considering that inversion introduces perturbations to latent coordinates, we adopt a k -truncated sampling strategy

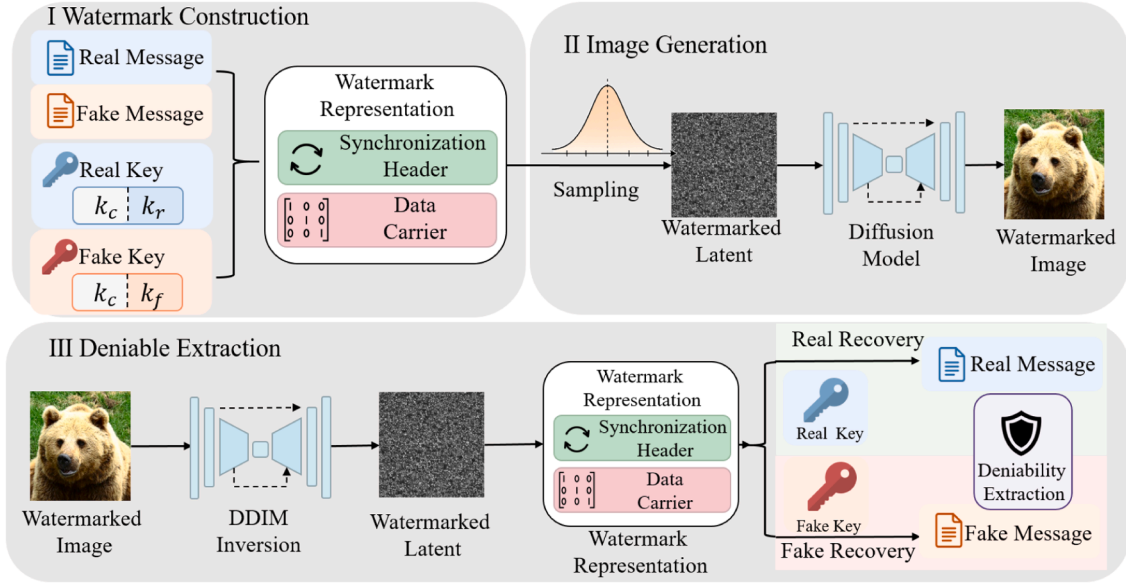


Fig. 2. Workflow of PDMark.

to prevent watermark bit flipping. Sampling driven by the watermark representation is restricted to the two intervals $(-\infty, -k]$ and $(k, +\infty)$, and the total probability mass of the two intervals is

$$p_k = 2(1 - \Phi(k)), \quad (17)$$

which determines the number of latent coordinates can be assigned to watermark-carrying positions.

For a representation of length M , we define the repetition factor as

$$q = \left\lfloor \frac{d p_k}{M} \right\rfloor, \quad (18)$$

which determines the repetition count for each watermark representation bit. Accordingly, the first Mq coordinates in Ω_v are partitioned into M disjoint representation groups:

$$J_i = \{\omega_{(i-1)q+1}, \dots, \omega_{iq}\}, \quad i = 1, \dots, M. \quad (19)$$

The synchronization groups $\{\mathcal{G}_\ell\}_{\ell=1}^{L_{\text{sync}}}$ are given by the first L_{sync} representation groups, and the data groups $\{\mathcal{J}_i^{\text{data}}\}_{i=1}^{M_{\text{data}}}$ are given by the remaining representation groups. The unused coordinates form the padding set $\mathcal{P}$.

To avoid deterministic repetition patterns, we further introduce a repeated-bit mask for each bit

$$b_\eta^{\text{wm}} = b_i \oplus r_\eta, \quad r_\eta = \text{PRNG}(k_c, v, \eta), \quad (20)$$

where $\eta = (i, t)$ indexes the t -th repeated coordinate of the i -th representation bit. This masking step prevents the q repeated copies of the same representation bit from forming a deterministic sign pattern and helps preserve the randomness of the embedded signs.

Let $\mathbf{y} \in \mathbb{R}^d$ denote the flattened latent to be sampled. For each representation bit b_η^{wm} and each repeated coordinate $\xi_{i,t} \in J_i$, we sample independently from a truncated standard Gaussian distribution according to the masked bit:

$$y_{\xi_{i,t}} \sim \begin{cases} \mathcal{N}(0, 1) \mid x > k, & b_\eta^{\text{wm}} = 1, \\ \mathcal{N}(0, 1) \mid x \leq -k, & b_\eta^{\text{wm}} = 0. \end{cases} \quad (21)$$

For all remaining coordinates, we sample from the central interval:

$$y_\xi \sim \mathcal{N}(0, 1) \mid -k < x \leq k, \quad \xi \in \mathcal{P}. \quad (22)$$

After sampling all coordinates, we reshape the sampled vector back to latent space and directly use it as the initial latent of the diffusion generator.

4.4. Deniable extraction

We first use the synchronization header to compensate for distribution drift and geometric distortions introduced during transmission, and then recover the watermark representation for message extraction.

Given a watermarked image $\tilde{\mathbf{x}}$, we map it back to latent space through DDIM inversion and flatten the recovered latent into $\mathbf{y}^{\text{ver}}$. Using the common component k_c and the public nonce v , we reconstruct the coordinate order and regenerate the synchronization component with Eqs. (16) and (10).

To compensate for geometric distortions, we use the synchronization header to evaluate candidate distortion parameters. Let θ denote the rotation angle and ϕ denote the flip mode. For each synchronization bit ℓ , let $\mathcal{G}_\ell$ be its repeated coordinate group, and let $r_\xi(\theta, \phi)$ be the recovered raw response at coordinate ξ under distortion parameters (θ, ϕ) . The synchronization matching score is defined as

$$S_{\text{sync}}(\theta, \phi) = \frac{1}{L_{\text{sync}}} \sum_{\ell=1}^{L_{\text{sync}}} \left[(2h_{v,\ell}^{\text{sync}} - 1) \cdot \frac{1}{|\mathcal{G}_\ell|} \sum_{\xi \in \mathcal{G}_\ell} r_\xi(\theta, \phi) \right], \quad (23)$$

where $h_{v,\ell}^{\text{sync}}$ is the ℓ -th bit of the key-derived synchronization component. The final geometric alignment is determined by

$$(\theta^*, \phi^*) = \arg \max_{\theta \in \Theta, \phi \in \Phi} S_{\text{sync}}(\theta, \phi). \quad (24)$$

After alignment, we estimate normalization parameters from the synchronization responses and normalize the recovered latent:

$$\bar{\mathbf{y}}^{\text{ver}} = \frac{\mathbf{y}^{\text{ver}} - \mu_v}{\sigma_v}, \quad (25)$$

where μ_v and σ_v are estimated using the recovered values on the synchronization groups and the known signs induced by $\mathbf{h}_v^{\text{sync}}$. These signs are available only to a holder of an extraction key containing k_c .

Next, we recover each representation bit by removing the repeated-bit mask and applying majority voting. For the i -th coordinate of the i -th representation group, we first obtain the raw repeated bit

$$\hat{b}_{i,t}^{\text{wm}} = \mathbb{I}(\bar{y}_{\xi_{i,t}}^{\text{ver}} > 0), \quad (26)$$

and then remove the mask derived from (k_c, v) .

The recovered representation bit is obtained by majority voting:

$$\hat{b}_i = \text{Maj}(\hat{b}_{i,1}, \dots, \hat{b}_{i,q}), \quad i = 1, \dots, M. \quad (27)$$

To decode a specific watermark $j \in \{r, f\}$, we regenerate the corresponding private relation from the private component with Eq. (9). The extracted encrypted string is computed only from the recovered data component:

$$\hat{\mathbf{c}}^{(j)} = (\mathbf{H}_v^{(j)} \hat{\mathbf{b}}^{\text{data}}) \oplus \mathbf{d}_v^{(j)}. \quad (28)$$

Finally, we decrypt the target watermark to obtain $\hat{\mathbf{m}}^{(j)}$. Specifically, the disclosed private key removes the same domain-separated pseudo-random stream:

$$\hat{\mathbf{m}}^{(j)} = \hat{\mathbf{c}}^{(j)} \oplus \mathcal{C}_{\text{enc}}(k_j, v; \text{msg-enc}). \quad (29)$$

5. Analysis

In this section, we analyze the proposed deniable watermarking framework from the perspectives of correctness, statistical indistinguishability, and security under the single-key coercive disclosure model.

5.1. Correctness analysis

We analyze the correctness of PDMark by considering whether the watermark representation can be recovered.

Let $\mathbf{y}$ be the sampled latent and $\mathbf{y}^{\text{ver}}$ be the recovered latent after generation and inversion. For the i -th representation bit, define the embedding group score as

$$\bar{r}_i = \frac{1}{q} \sum_{t=1}^q (2b_{i,t}^{\text{wm}} - 1) |y_{\bar{\epsilon}_{i,t}}|, \quad (30)$$

where $b_{i,t}^{\text{wm}} = b_i \oplus r_{i,t}$ is the masked repeated bit. After synchronization correction and mask removal, let the recovered group score be

$$\hat{r}_i = \bar{r}_i + \bar{\epsilon}_i, \quad (31)$$

where $\bar{\epsilon}_i$ denotes the group-level perturbation caused by inversion and distortions.

Proposition 1 (Correctness). *Let $\mathbf{b}$ be the embedded watermark representation and $\hat{\mathbf{b}}$ be the recovered representation. If the group-level perturbation satisfies*

$$|\bar{\epsilon}_i| < |\bar{r}_i|, \quad i = 1, \dots, M, \quad (32)$$

then $\hat{\mathbf{b}} = \mathbf{b}$. Consequently, the watermark decoded under the corresponding extraction key is recovered correctly.

Proof. The extraction of each representation bit depends on the sign of its recovered group score. For the i -th group, the recovered score can be written as

$$\hat{r}_i = \bar{r}_i + \bar{\epsilon}_i, \quad (33)$$

where $\bar{r}_i$ is the embedding group score and $\bar{\epsilon}_i$ denotes the group-level perturbation introduced by inversion and distortions. Xu et al. (2025b) has observed that the element-wise inversion error of the recovered initial noise is usually bounded within a relatively small range in latent diffusion models. Motivated by this observation, our sampling strategy places watermark-carrying latent values in the truncated tail intervals $(-\infty, -k]$ and $(k, +\infty)$, so that each embedded bit has a non-negligible sign margin from zero. This margin reduces the probability that inversion perturbations flip the sign used for bit recovery.

Under $|\bar{\epsilon}_i| < |\bar{r}_i|$, the perturbation cannot flip the sign of $\bar{r}_i$, so $\hat{r}_i$ preserves the embedded sign. After removing the repeated-bit mask derived from (k_c, v) , each bit is recovered correctly as $\hat{\mathbf{b}} = \mathbf{b}$. $\square$

5.2. Statistical indistinguishability

We analyze the statistical indistinguishability of the watermarked latent from the perspective of Gaussian consistency. Sampling is performed over three disjoint Gaussian regions, i.e. $(-\infty, -k]$, $(-k, k]$, and $(k, +\infty)$. Watermark-carrying coordinates are sampled from the two tails according to the masked representation bits, while padding coordinates are sampled from the central interval. Let S_v denote the set of watermark-carrying coordinates, with $|S_v| = Mq$. For any coordinate, the probability of being selected is

$$\rho = \Pr[a \in S_v] = \frac{Mq}{d}. \quad (34)$$

Since the encrypted representation bits and repeated-bit masks make the embedded signs computationally indistinguishable from unbiased Bernoulli bits, a selected coordinate is assigned to the positive and negative tails with equal probability. Therefore, the marginal density of an arbitrary latent coordinate can be written as

$$\begin{aligned} p(y) &= \frac{\rho}{2} p_{-}(y) + (1 - \rho) p_0(y) + \frac{\rho}{2} p_{+}(y) \\ &= \Phi(-k) p_{-}(y) + (\Phi(k) - \Phi(-k)) p_0(y) \\ &\quad + (1 - \Phi(k)) p_{+}(y) = \phi(y). \end{aligned} \quad (35)$$

where p_{-} , p_0 , and p_{+} are the standard Gaussian distributions truncated to the negative tail, central region, and positive tail, respectively. So the watermarked latent is statistically indistinguishable from a standard Gaussian latent.

5.3. Security analysis

We now analyze the security objectives of the proposed framework under the single-key coercive disclosure model. Let $\tau \in \{r, f\}$ denote the disclosed watermark and let $\bar{\tau}$ denote the undisclosed watermark in the following.

Pre-disclosure coordinate hiding. Before any extraction key is disclosed, the adversary does not know k_c . The coordinate order, synchronization positions, and repeated-bit masks are generated by pseudorandom functions keyed by k_c . Therefore the adversary cannot identify which latent coordinates are used for watermark-carrying groups or synchronization groups beyond random guessing. This property concerns the hidden coordinate structure before disclosure, while image-level key-less undetectability is further supported by the Gaussian consistency.

Proposition 2 (Coordinate Hiding). *Assume that $\mathcal{G}_\Omega$ and the pseudo random generators are secure. Before any extraction key is disclosed, no efficient adversary can identify the watermark-carrying coordinate groups or synchronization positions with non-negligible advantage.*

Proof. The coordinate order and synchronization component are generated from the common component k_c and the public nonce v . Let

$$S_v \approx_c S_v^{\text{rand}}, \quad (36)$$

where $S_v = (\Omega_v, \mathbf{h}_v^{\text{sync}}, \mathbf{R}_v)$, and S_v^{rand} denotes a structure consisting of a uniformly random permutation and random binary patterns of the same dimensions. This computational indistinguishability is guaranteed by the pseudo randomness assumption on $\mathcal{G}_\Omega$ and the associated PRNGs. Suppose an efficient adversary could identify the watermark-carrying groups or synchronization positions with non-negligible advantage before any extraction key is disclosed. Then its success probability would differ non-negligibly between the real structure and the random structure:

$$|\Pr[\text{Succ}_{\mathcal{A}}(S_v)] - \Pr[\text{Succ}_{\mathcal{A}}(S_v^{\text{rand}})]| > \text{negl}(\lambda). \quad (37)$$

Such an adversary would directly distinguish the generator outputs from random outputs, violating the pseudo randomness assumption. Therefore, before any extraction key is disclosed, the coordinate structure remains computationally hidden. $\square$

Content confidentiality. We next consider the confidentiality of the undisclosed watermark after one extraction key has been disclosed. The disclosed key may reveal the common structure and the disclosed watermark, but it should not reveal the content of the undisclosed watermark. Even if the adversary reconstructs the coordinate groups and recovers the watermark representation, decoding the undisclosed watermark still requires the private component $k_{\bar{r}}$. Without $k_{\bar{r}}$, the adversary cannot reconstruct $(\mathbf{H}_v^{(\bar{r})}, \mathbf{d}_v^{(\bar{r})})$.

Proposition 3 (Content Confidentiality). *Assume that $\mathcal{G}_H$ is pseudorandom and that encrypted watermark strings are computationally indistinguishable from random bit strings. Disclosure of one extraction key does not reveal the content of the undisclosed watermark without $k_{\bar{r}}$.*

Proof. After the extraction key $k^{(r)} = (k_c, k_r)$ is disclosed, the adversary may know the common component k_c and the disclosed private component k_r . Thus, the adversary can reconstruct the coordinate assignment, the synchronization pattern, and the disclosed relation. For the undisclosed watermark, extraction requires $(\mathbf{H}_v^{(\bar{r})}, \mathbf{d}_v^{(\bar{r})})$ and the corresponding watermark string is computed as

$$\hat{\mathbf{c}}^{(\bar{r})} = (\mathbf{H}_v^{(\bar{r})} \hat{\mathbf{b}}^{\text{data}}) \oplus \mathbf{d}_v^{(\bar{r})}. \quad (38)$$

Since $k_{\bar{r}}$ is unknown, $(\mathbf{H}_v^{(\bar{r})}, \mathbf{d}_v^{(\bar{r})})$ is computationally indistinguishable from a random pair $(\mathbf{R}, \mathbf{s})$ of the same size. Therefore,

$$\hat{\mathbf{c}}^{(\bar{r})} \approx_c (\mathbf{R} \hat{\mathbf{b}}^{\text{data}}) \oplus \mathbf{s}. \quad (39)$$

For any fixed recovered representation $\hat{\mathbf{b}}^{\text{data}}$, the value $(\mathbf{R} \hat{\mathbf{b}}^{\text{data}}) \oplus \mathbf{s}$ is computationally indistinguishable from a random bit string because $\mathbf{s}$ is random. Hence, the adversary obtains no useful information about the encrypted string of the undisclosed branch. Since the encrypted watermark string is also assumed to be indistinguishable from random without the corresponding private information, the adversary cannot recover the undisclosed watermark content with non-negligible advantage. $\square$

Disclosure ambiguity. Content confidentiality only ensures that the undisclosed watermark cannot be recovered. Deniability further requires that the disclosed watermark itself does not reveal whether it corresponds to the real one or the fake one. In PDMark, both watermark are extracted through the same pipeline and thus the algorithmic extraction process is symmetric with respect to the two watermarks.

Proposition 4 (Disclosure Ambiguity). *If the real and fake watermark messages follow the same distribution, then the disclosed watermark does not reveal its role.*

Proof. Let $T \in \{r, f\}$ denote the hidden role of the disclosed watermark. Here, $T = r$ means that the disclosed watermark corresponds to the real one, and $T = f$ means that it corresponds to the fake one. Let $\hat{\mathbf{m}}$ denote the extracted watermark obtained from the disclosed extraction key. The coordinate groups, synchronization adjustment, repeated-bit aggregation, and representation recovery of the two watermarks are identical once the common component is known. The only different operation is the final decoding relation derived from the corresponding private component. Therefore, conditioned on successful extraction, the adversary observes a message in the same message space for both watermarks, and their distributions are indistinguishable:

$$\Pr[\hat{\mathbf{m}} = \mathbf{m} \mid T = r] \approx \Pr[\hat{\mathbf{m}} = \mathbf{m} \mid T = f], \quad \forall \mathbf{m}. \quad (40)$$

For any efficient distinguisher $\mathcal{A}$ that attempts to decide whether the disclosed watermark is real or fake, its advantage is bounded by the statistical distance between these two message distributions. Under the same-distribution assumption, this distance is negligible. Hence, the disclosed watermark provides no non-negligible evidence about whether its role is real or fake. $\square$

Post-disclosure existence ambiguity. After one extraction key is disclosed, the adversary can reconstruct the common coordinate structure and verify the disclosed watermark. The stronger requirement is that such knowledge should not allow the adversary to determine whether the undisclosed one carries another meaningful watermark or only auxiliary random padding.

We formalize this property through a post-disclosure distinguishing game. Let $\tau \in \{r, f\}$ denote the disclosed watermark and let $\bar{\tau}$ denote the undisclosed watermark. In both hypotheses, the disclosed watermark τ is constructed, embedded, and extracted in the same way. The adversary obtains the watermarked image, the public nonce, the disclosed extraction key and the corresponding extracted watermark. The adversary does not receive $k_{\bar{r}}$. The two hypotheses are defined as:

- $\mathcal{H}_0$: **Auxiliary-padding hypothesis.** The undisclosed watermark has the same structure but carries an auxiliary random payload.
- $\mathcal{H}_1$: **Meaningful-watermark hypothesis.** The undisclosed watermark has the same structure but carries another encrypted meaningful message.

Post-disclosure existence ambiguity requires that no efficient adversary can distinguish these two hypotheses with non-negligible advantage.

Proposition 5 (Existence Ambiguity). *Assume that encryption outputs are indistinguishable from random, $\mathcal{G}_H$ is pseudorandom, and $\mathbf{b}^{\text{data}}$ is sampled uniformly from the corresponding space. Then the adversary cannot distinguish whether the undisclosed watermark carries a meaningful message or a same-form auxiliary random payload.*

Proof. We compare the adversary's observations under $\mathcal{H}_0$ and $\mathcal{H}_1$. In both hypotheses, the disclosed watermark τ imposes the same visible constraint on the data representation:

$$\mathbf{H}_v^{(\tau)} \mathbf{b}^{\text{data}} = \hat{\mathbf{c}}^{(\tau)} \oplus \mathbf{d}_v^{(\tau)}. \quad (41)$$

Since the disclosed key contains k_c and k_r , the adversary can reconstruct and verify the disclosed watermark in both hypotheses, providing identical verification evidence under $\mathcal{H}_0$ and $\mathcal{H}_1$. The only difference lies in the undisclosed branch. Under both hypotheses, the undisclosed watermark has the same structure:

$$\mathbf{H}_v^{(\bar{\tau})} \mathbf{b}^{\text{data}} = \mathbf{s}_v^{(\bar{\tau})}. \quad (42)$$

Under the meaningful-watermark hypothesis $\mathcal{H}_1$, the right-side is

$$\mathbf{s}_v^{(\bar{\tau})} = \hat{\mathbf{c}}^{(\bar{\tau})} \oplus \mathbf{d}_v^{(\bar{\tau})}, \quad (43)$$

where $\hat{\mathbf{c}}^{(\bar{\tau})}$ is the encrypted form of another meaningful watermark. Under the auxiliary-padding hypothesis $\mathcal{H}_0$, the right-side is

$$\mathbf{s}_v^{(\bar{\tau})} = \mathbf{u}_v^{(\bar{\tau})}, \quad \mathbf{u}_v^{(\bar{\tau})} \leftarrow \{0, 1\}^L. \quad (44)$$

The two right-sides have the same length and format. We use a hybrid argument. First, under $\mathcal{H}_1$, replace the encrypted watermark string $\hat{\mathbf{c}}^{(\bar{\tau})}$ with a uniformly random string of the same length. By the assumed indistinguishability of encryption outputs, no efficient adversary can detect this replacement with non-negligible advantage. After this replacement, $\hat{\mathbf{c}}^{(\bar{\tau})} \oplus \mathbf{d}_v^{(\bar{\tau})}$ is computationally indistinguishable from a random string of length L . Second, since $k_{\bar{r}}$ is not disclosed, the hidden relation $(\mathbf{H}_v^{(\bar{\tau})}, \mathbf{d}_v^{(\bar{\tau})})$ is computationally indistinguishable from a random relation and a random offset of the same dimensions.

After marginalizing over the undisclosed private component, the hidden branch under $\mathcal{H}_1$ has the same distribution as the auxiliary branch under $\mathcal{H}_0$: both are same-form hidden constraints with random targets. Since the data representation is uniformly sampled from the corresponding space in both hypotheses, the recovered representation and the disclosed watermark are computationally indistinguishable between $\mathcal{H}_0$ and $\mathcal{H}_1$. Thus, any efficient distinguisher would violate the assumed encryption indistinguishability or the pseudo randomness of $\mathcal{G}_H$. Therefore, the undisclosed watermark is computationally indistinguishable from a same-form auxiliary random branch. $\square$

5.4. Complexity analysis

Let $d = C \times H \times W$ be the latent dimension, M_{data} the length of the data component, and L the message length of each branch.

During watermark construction, the main cost is solving the shared binary system in Eq. (12). Using a standard elimination-based solver, this step costs $O(L^2 M_{\text{data}})$. The latent sampling stage generates the coordinate order once and samples each latent coordinate at most once, leading to an embedding overhead of $O(d)$.

During extraction, synchronization scoring and representation recovery are linear in the latent dimension. Decoding one watermark further requires a binary matrix-vector multiplication with cost $O(L M_{\text{data}})$. Thus, the watermark-specific extraction cost is $O(d + L M_{\text{data}})$.

The full image-generation or inversion pipeline still costs approximately $O(T C_{\text{UNet}})$ for T DDIM steps and per-step UNet cost C_{UNet} .

6. Experiments

6.1. Experimental setup

Models and datasets. We conduct experiments on Stable Diffusion v2.1 by default. We use the image-prompt pairs from COCO 2017 as the validation prompts, and use the corresponding images to evaluate the fidelity of the proposed method. Imperceptibility and robustness evaluations are tested on 1000 image-prompt pairs.

Implementation details. All experiments were conducted with a single NVIDIA GeForce RTX 5090 GPU and an Intel Xeon Platinum 8358P CPU at 2.60 GHz. The latent size is fixed to $d = 4 \times 64 \times 64$, and we set the number of representation coordinates to $M = 512$. Both the real watermark and the fake watermark are 128 bits, and the sample truncation threshold is set to $k = 1.15$. The public nonce is stored with the generated image metadata and can be used to regenerate the per-image public randomness.

For diffusion generation, we use 50 sampling steps by default. In conditional generation, the guidance scale is fixed to 7.5, following the implementation used throughout evaluation, while DDIM inversion is performed under a null-prompt diffusion process unless otherwise specified.

For attacks applied in image space, the attacked image is first inverted back to the latent space, and the recovered latent is then used for representation recovery and message decoding. For rotation attacks, the image is rotated in image space with bicubic interpolation while keeping the output size unchanged. For scale attacks, the image is resized by the specified scale factor and then restored to the original resolution before extraction.

Metrics. We evaluate the proposed method in terms of extraction accuracy, deniability, robustness, and model fidelity.

For extraction accuracy and robustness, we use bit accuracy (ACC) as the extraction metric, defined as the proportion of correctly recovered watermark bits among all embedded bits. We evaluate robustness under three kinds of perturbations: *conventional signal-processing attacks (CSP)*, including JPEG compression, Gaussian noise, Gaussian blur, median filtering, and brightness adjustment; *geometric attacks*, including rotation, flip and scale; and *regeneration and purification attacks*. For ease of comparison with the baselines, and considering that the two deniable watermarks exhibit consistent performance, we use the average ACC of the two watermarks as the final performance metric.

For deniability, we additionally evaluate extraction performance under random guessed keys and coercive disclosure. A deniable watermarking method is expected to maintain high ACC under the corresponding key while producing random outputs under incorrect keys. Moreover, under coercive disclosure, the extracted watermark revealed by a key should remain ambiguous without revealing the existence of the other watermark.

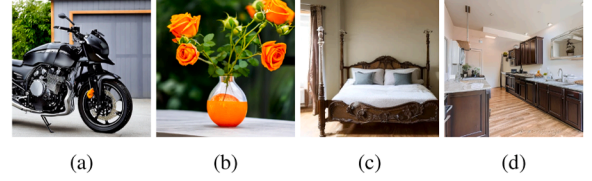


Fig. 3. Qualitative examples of watermarked images. *Prompts.* (a) A black Honda motorcycle parked in front of a garage; (b) An orange reddish rose in a vase with water on top of a table; (c) An old victorian style bed frame in a bedroom; (d) Kitchen area with counter top space and dining room table.

For model fidelity, we report Fréchet Inception Distance (FID) and CLIP Score to assess image quality. FID measures the distributional difference between generated images and reference images in the feature space of a pretrained Inception network. CLIP Score measures the semantic consistency between a generated image and its prompt using a pretrained vision-language model.

6.2. Baselines

We compare our method with representative diffusion watermarking baselines to demonstrate that introducing deniability does not sacrifice the basic performance of watermarking, and that our method remains competitive with SOTA methods while supporting deniable extraction. Baselines contain both categories: model-tuning-based methods, including *Stable Signature* (Fernandez et al., 2023), *WaDiff* (Min et al., 2024), *AquaLoRA* (Feng et al., 2024), and *LaWa* (Rezaei et al., 2024); and latent-based methods, including *Tree-Ring Watermarks* (Wen et al., 2023), *RingID* (Ci et al., 2024), *ROBIN* (Huang et al., 2024), *WIND* (Arabi et al., 2025), and *Gaussian Shading* (Yang et al., 2024). For extraction performance, we report bit accuracy (ACC) for multi-bit methods and true positive rate (TPR) for 1-bit methods. Multi-bit extraction is a payload-recovery task, for which bitwise ACC directly measures the proportion of correctly recovered bits, whereas 1-bit extraction is a watermark-presence detection task, for which TPR directly measures the proportion of watermarked images correctly detected and is consistent with the evaluation protocols of the original baselines. By comparing with baselines, we demonstrate that deniability does not compromise the fundamental performance and robustness of the watermark.

6.3. Qualitative results

To provide a direct visual assessment of the proposed method, we present representative watermarked images in Fig. 3. The examples show that the generated images preserve normal visual appearance and semantic consistency with the prompts, indicating that the proposed latent watermark construction does not introduce obvious visible artifacts.

6.4. Deniability evaluation

Deniability is the central security property of our method. We evaluate PDMark from four aspects in the threat model: *pre-disclosure undetectability*, where Carol should not identify the watermark or distinguish watermarked images from ordinary generated images without a disclosed key; *content confidentiality*, where disclosing one extraction key should not reveal the content of the undisclosed watermark; *disclosure ambiguity*, where Carol should not determine whether the extracted watermark is real or fake; and *post-disclosure existence ambiguity*, where Carol should not distinguish whether the undisclosed watermark carries meaningful information or auxiliary random padding. Accordingly, we conduct pre-disclosure undetectability evaluation, content confidentiality evaluation, and post-disclosure ambiguity evaluation.

Table 1
Pre-disclosure undetectability evaluation.

Level	Detector	ACC	AUC	TPR@1%FPR
Latent	Tail count	0.521	0.518	0.022
Latent	KS statistic	0.516	0.523	0.014
Latent	AD statistic	0.528	0.536	0.017
Image	CLIP feat. + LR	0.506	0.508	0.011
Image	CLIP feat. + SVM	0.512	0.515	0.013

Table 2
Extraction results under different key selections.

Image	Key selection	Target watermark	ACC
Watermarked	$k^{(r)}$	$\mathbf{m}^{(r)}$	1.000
	$k^{(f)}$	$\mathbf{m}^{(f)}$	1.000
	$k^{(f)}$	$\mathbf{m}^{(r)}$	0.498
	$k^{(r)}$	$\mathbf{m}^{(f)}$	0.493
	Random	$\mathbf{m}^{(r)}$	0.494
	Random	$\mathbf{m}^{(f)}$	0.493
Clean	$k^{(r)}$	$\mathbf{m}^{(r)}$	0.497
	$k^{(f)}$	$\mathbf{m}^{(f)}$	0.495

Table 3
Wrong-private key attack.

Image	Key selection	Target Watermark	ACC
Watermarked	(k_c, k_{wrong})	$\mathbf{m}^{(r)}$	0.493
Watermarked	(k_c, k_{wrong})	$\mathbf{m}^{(f)}$	0.506
Clean	(k_c, k_{wrong})	$\mathbf{m}^{(r)}$	0.501
Clean	(k_c, k_{wrong})	$\mathbf{m}^{(f)}$	0.508

Pre-disclosure undetectability evaluation. Before any extraction key is disclosed, the adversary may attempt to distinguish PDMark images from ordinary generated images. We compare ordinary images generated from standard Gaussian latents with PDMark images generated from the same prompt set.

To evaluate possible artifacts, we consider both latent-level and image-level detectors. At the latent level, we apply tail-count, KS, and Anderson–Darling statistics to DDIM-inverted latents. At the image level, we train classifiers based on CLIP features, including logistic regression (LR) and support vector machine (SVM), to distinguish ordinary generated images from PDMark images. As shown in Table 1, all evaluated detectors remain close to random guessing.

Content confidentiality evaluation. Content confidentiality requires that disclosing one extraction key should not enable Carol to recover meaningful information about the undisclosed watermark.

We first verify the basic key-selective behavior of the proposed method. As shown in Table 2, the corresponding key successfully recovers the intended watermark, whereas mismatched keys and randomly guessed keys only produce nearly random outputs with ACC close to 0.5. Moreover, when the valid keys are applied to clean images without watermark, the recovered bits also remain near random. These results verify the key-selective behavior of PDMark and show that meaningful recovery requires the matching extraction key.

The mismatched-key setting above considers random guessed keys. Under coercive disclosure, however, the adversary may already know the common component k_c from the disclosed extraction key and only needs to guess the undisclosed private component. We therefore evaluate a wrong-private key attack, where the adversary uses a key of the form (k_c, k_{wrong}) to reconstruct the common coordinate structure, remove the repeated-bit masks, and attempt to decode the undisclosed watermark with an incorrect private component. As shown in Table 3, the extracted results remain close to random guessing, indicating that knowing the common component alone does not enable meaningful recovery of the undisclosed watermark.

Table 4
Post-disclosure ambiguity evaluation.

Task	Detector	ACC	AUC	EER
Disclosure Ambiguity	Heuristic	0.510	0.498	0.503
	Learned	0.526	0.512	0.494
Existence Ambiguity	Logistic reg.	0.503	0.497	0.491
	Random forest	0.509	0.500	0.495
	MLP	0.506	0.498	0.504

Post-disclosure evaluation. We evaluate deniability under coercive disclosure, in other words, the last two security objectives in the threat model.

For disclosure ambiguity, we disclose either the real key or the fake key, and test whether the adversary can determine which watermark has been revealed. We consider a heuristic detector based on the average absolute log-likelihood ratio $\bar{\ell}_{\text{abs}} = \frac{1}{L} \sum_{i=1}^L |\ell_i|$, where ℓ_i denotes the bit-level log-likelihood ratio, and a learned detector implemented as a linear probe over image features and disclosure statistics. The learned detector considers LLR confidence, raw response statistics and byte entropy, and then trains a binary linear classifier to distinguish real-key disclosure from fake-key disclosure. As shown in the first part of Table 4, both detectors perform close to random guessing, indicating that once a valid key is disclosed, the adversary cannot reliably determine whether the disclosed result corresponds to the real watermark or the fake watermark.

We further evaluate the post-disclosure existence ambiguity. For each disclosed watermark, we construct two hypotheses: under $\mathcal{H}_0$, the undisclosed watermark is assigned an auxiliary random payload of the same length and format; under $\mathcal{H}_1$, the undisclosed watermark carries another meaningful encrypted message. The adversary obtains the disclosed key $k^{(r)} = (k_c, k_r)$, and all public hyperparameters, and can therefore reconstruct the data representation $\hat{\mathbf{b}}^{\text{data}}$.

We train three lightweight detectors, including logistic regression, random forest, and MLP, to distinguish $\mathcal{H}_0$ from $\mathcal{H}_1$ using post-disclosure statistics derived from the recovered representation and decoding process. As shown in the second part of Table 4, their accuracies and AUC values remain close to random guessing, indicating that the undisclosed watermark is not reliably distinguished from auxiliary random padding under the tested post-disclosure detectors.

6.5. Known-plaintext security evaluation

In this known-plaintext security evaluation, we examine whether the disclosed key–watermark pair leaks information about the hidden branch under the strong post-disclosure threat model. We consider two complementary objectives: direct recovery of the hidden ciphertext or plaintext from the disclosed information, and detection of residual hidden-branch correlations through high-order statistical attacks.

Known-plaintext attacks. The attacker receives the public nonce, disclosed key, disclosed plaintext, and all public parameters. We consider a *same-key* protocol, which uses the same key identities but disjoint samples and nonces across data splits, and a *cross-key* protocol, which evaluates generalization to unseen keys.

Table 5 reports four complementary evaluations. The *nonce-only control* tests whether the public nonce alone leaks information about the hidden branch. The *hidden-ciphertext recovery* and *hidden-plaintext recovery* attacks attempt to reconstruct the 128-bit hidden ciphertext and its corresponding plaintext watermark, respectively. The *wrong-private-key control* tests whether the disclosed common key combined with an incorrect private key enables meaningful extraction. We report AUC for nonce-only classification and bit ACC for the other evaluations, with values close to 0.5 indicating random guessing.

The nonce-only control uses five construction seeds, 1000 samples per class, and 20 keys. The recovery experiments use 20 construction

Table 5
Known-plaintext attack results.

Evaluation	Metric	Same-key	Cross-key
Nonce-only control	AUC	0.5005	0.4981
Hidden-ciphertext recovery	ACC	0.4998	0.4996
Hidden-plaintext recovery	ACC	0.5002	0.4997
Wrong-private-key control	ACC	0.5001	0.5000

Table 6
AUC results for high-order statistical attacks.

Detector	Carrier only	Carrier with nonce	Min. FDR q
ExtraTrees	0.4989	0.4958	0.9305
Logistic regression	0.4976	0.4949	0.9305
MLP	0.5059	0.4960	0.3439
RBF-SVM	0.4980	0.4992	0.9305

seeds, 2000 records per seed, 20 keys, and disjoint 1,100/300/600 training, validation, and test partitions, resulting in 160 attacks. Each recovery task is evaluated both with and without the public nonce.

For hidden-ciphertext and hidden-plaintext recovery, we train a supervised multi-output MLP with two hidden layers of 256 and 128 units and 128 output logits. Its standardized input combines carrier and disclosed-coset features derivable from the disclosed key–plaintext pair with statistical moments, autocorrelations, random projections, degree-2 to degree-4 sketches, and optional nonce features. The model is trained using AdamW and binary cross-entropy.

As shown in Table 5, all evaluations remain close to random guessing under both protocols. None of the 160 recovery attacks reconstructs a complete 128-bit hidden message, indicating that the disclosed information does not enable practical recovery of the hidden branch.

High-order statistical attacks. Direct recovery attacks may fail even when the carrier contains weak but detectable statistical correlations. We therefore further examine whether the structured linear relations leave a distinguishable hidden-branch signal. The attacker evaluates both *carrier-only* and *carrier-with-nonce* feature views using statistical moments, autocorrelations, random projections, and degree-2 to degree-4 monomial sketches. Logistic regression, RBF-SVM, ExtraTrees, and MLP detectors are used to cover linear, kernel-based, tree-based, and neural attacks, respectively.

The evaluation uses 20 independent construction seeds, 1000 samples per class, and 20 keys, resulting in 480 attacks. We report mean AUC and the minimum Benjamini–Hochberg-corrected FDR q -value across the two feature views.

As shown in Table 6, all detectors remain close to random guessing, and including the nonce provides no consistent improvement. The meaningful-versus-auxiliary and cross-key dual-versus-single controls also yield chance-level results, providing no evidence of a stable or transferable hidden-branch signal.

Overall, neither direct recovery nor high-order statistical detection reveals practically exploitable information about the hidden branch. These results provide empirical evidence under the evaluated attack suite rather than an unconditional security guarantee.

6.6. Comparative experiment

Basic performance comparison. We compare the proposed method with state-of-the-art (SOTA) diffusion watermarking baselines without distortions.

Table 7 compares the proposed method with representative baselines in terms of payload length, extraction performance, image fidelity, and semantic consistency. Although existing baselines are designed for deterministic extraction and use different payload settings, they provide a useful reference for evaluating the standard watermarking behavior of our method. PDMark achieves perfect clean extrac-

Table 7
Basic performance comparison.

Method	Length (bits)	Metric	Accuracy $\uparrow$	FID $\downarrow$	CLIP $\uparrow$
Model-tuning-based					
Stable	48	ACC	0.994	25.45	0.362
Signature (Fernandez et al., 2023)					
WaDiff (Min et al., 2024)	48	ACC	0.982	25.07	0.315
AquaLoRA (Feng et al., 2024)	48	ACC	0.958	24.88	0.311
LaWa (Rezaei et al., 2024)	48	ACC	1.000	25.42	0.366
Latent-based					
Tree-Ring (Wen et al., 2023)	1	TPR	1.000	25.43	0.365
ROBIN (Huang et al., 2024)	1	TPR	0.983	26.86	0.396
WIND (Arabi et al., 2025)	1	TPR	0.971	24.33	0.360
RingID (Ci et al., 2024)	11	ACC	1.000	25.43	0.366
Gaussian Shading (Yang et al., 2024)	256	ACC	1.000	25.20	0.364
Ours	2×128	ACC	1.000	25.26	0.367

Table 8
Runtime overhead comparison.

Method	Latent construction	Watermark extraction
Tree-Ring (Wen et al., 2023)	0.3619 ms	0.2241 ms
RingID (Ci et al., 2024)	0.7476 ms	1.5698 ms
ROBIN (Huang et al., 2024)	0.7075 ms	0.5274 ms
WIND (Arabi et al., 2025)	0.4843 ms	2.0457 ms
Gaussian Shading (Yang et al., 2024)	2096 ms	0.4628 ms
PDMark	66.7592 ms	0.9190 ms

tion accuracy with a 2×128 -bit deniable payload, while maintaining FID and CLIP scores comparable to prior diffusion watermarking methods. This indicates that introducing deniable extraction does not noticeably compromise visual quality or text-image alignment under the clean setting.

Runtime overhead comparison. We compare watermarked-latent construction and watermark extraction with latent-based methods, respectively. All methods are evaluated with batch size 1 and latent size $4 \times 64 \times 64$. Each value in Table 8 is averaged over 1000 timed run.

PDMark requires more construction time than single-watermark methods because it additionally performs deniable dual-branch carrier construction. Nevertheless, its construction time of 66.7592 ms remains substantially lower than the 2096 ms required by Gaussian Shading. DDIM generation and inversion stages require approximately 1.7 s and 1.2 s per image, respectively. Therefore, PDMark’s watermarked-latent construction accounts for only about 2.3% of their combined runtime, showing that the additional deniable computation introduces a relatively small overhead to the complete generation and verification pipeline.

Robustness comparison. We further compare robustness under various attacks, including CSP attacks and geometric attacks. We apply Gaussian noise with $\sigma = 8$, Gaussian blur with radius $r = 4$, median filtering with kernel size 7, JPEG compression with quality factor $Q = 60$, rotation by 45° , horizontal flipping, scaling with factor 0.75, and brightness adjustment with factor 1.5. As shown in Table 9, PDMark maintains high average extraction accuracy under most signal-processing and geometric distortions.

6.7. Robustness against regeneration and purification

Table 10 presents the robustness of our method under four representative content-restoration attacks, including three regeneration-based

Table 9
Robustness comparison.

Method	Gaussian Noise	Gaussian Blur	Median Filter	JPEG	Flip	Scale	Rotation	Brightness
Model-tuning-based								
Stable Signature (Fernandez et al., 2023)	0.789	0.899	0.940	0.880	0.448	0.910	0.591	0.968
WaDiff (Min et al., 2024)	0.725	0.874	0.784	0.803	0.547	0.895	0.606	0.845
AquaLoRA (Feng et al., 2024)	0.930	0.959	0.885	0.949	0.526	0.914	0.450	0.934
LaWa (Rezaei et al., 2024)	0.897	0.956	0.932	0.942	0.604	0.924	0.558	0.981
Latent-based								
Tree-Ring (Wen et al., 2023)	0.945	0.571	0.583	0.553	0.644	0.961	0.796	0.942
Gaussian Shading (Yang et al., 2024)	1.000	0.985	1.000	1.000	0.506	1.000	0.513	0.996
RingID (Ci et al., 2024)	0.946	0.981	0.934	0.986	1.000	1.000	0.871	0.894
ROBIN (Huang et al., 2024)	0.954	0.999	0.982	0.999	0.801	0.994	0.957	0.975
WIND (Arabi et al., 2025)	0.980	1.000	0.972	1.000	0.842	0.940	0.930	0.980
Ours	0.988	0.974	0.997	0.989	0.997	0.995	0.994	0.997

Table 10
Accuracy performance under regeneration and purification attacks.

Regen Diffusion	Step = 20	Step = 40	Step = 60	Step = 80
ACC Real	1.0000	0.9967	0.9948	0.9905
ACC Fake	0.9986	0.9930	0.9869	0.9823
ACC avg	0.9993	0.9949	0.9909	0.9864
Regen VAE	Quality = 1	Quality = 2	Quality = 3	Quality = 4
ACC Real	0.8780	0.9224	0.9576	0.9715
ACC Fake	0.8672	0.9014	0.9348	0.9581
ACC avg	0.8726	0.9119	0.9462	0.9648
Rinse	Step = 1	Step = 2	Step = 3	Step = 4
ACC Real	0.9984	0.9942	0.9823	0.9645
ACC Fake	0.9931	0.9847	0.9670	0.9422
ACC avg	0.9958	0.9895	0.9747	0.9534
Purification	Step = 10	Step = 20	Step = 30	Step = 40
ACC Real	0.9944	0.9875	0.9763	0.9594
ACC Fake	0.9875	0.9747	0.9570	0.9320
ACC avg	0.9910	0.9811	0.9667	0.9457

attacks, Regen-Diff (Meng et al., 2022), Regen-VAE (Ballé et al., 2018), and Rinse (An et al., 2024), and the purification attack DiffPure (Nie et al., 2022). Specifically, Regen-Diff performs a single diffusion-based noising-denoising regeneration, Regen-VAE removes watermark traces through compression and reconstruction with a pretrained variational image codec, Rinse strengthens watermark removal by repeatedly applying diffusion regeneration. Purification first perturbs the input through a forward diffusion process and then restores it via reverse denoising. Despite the substantial content perturbations introduced by these attacks, our method maintains strong robustness across all four settings.

6.8. Cross-model generalization

Although Stable Diffusion v2.1 is used as the default model in most experiments, the proposed method is designed as a train-free latent watermarking framework and does not rely on model-specific watermark training. We therefore evaluate its clean cross-model transferability across different Stable Diffusion variants. Specifically, we consider Stable Diffusion v1.4, v2.0, and v2.1 as both generation and extraction models.

As shown in Table 11, the proposed method maintains consistently high extraction accuracy across different combinations. This result suggests that the sign-based latent representation and the corresponding key-controlled recovery procedure are not tightly coupled to a single Stable Diffusion checkpoint. The reason for strong transfer performance is that the watermark is embedded in the initial latent representation, without requiring model-specific decoder or fine-tuned watermark extractor.

Table 11
Cross-model transfer performance.

Generation model	Extraction model		
	SD v1.4	SD v2.0	SD v2.1
SD v1.4	0.997	0.998	0.998
SD v2.0	0.998	0.999	1.000
SD v2.1	0.999	0.998	1.000

6.9. Ablation studies

We conduct ablation studies on the components and parameter choices of our framework to validate the effectiveness of the experimental design in terms of robustness, fidelity, and deniability.

The truncation threshold k jointly controls the overall embedding capacity and the repetition factor of the watermark bits. If k is too small, the embedded bits become more vulnerable to sign flipping caused by inversion errors. If k is too large, the available latent region that can accommodate watermark embedding while preserving Gaussian marginal consistency becomes much smaller. We therefore perform an ablation study on k , as reported in the first part of Table 12.

Table 12 further studies the effect of the watermark length M on extraction robustness in the Second part. We first observe that all tested settings maintain nearly perfect performance under the clean condition, indicating that changing M does not materially affect extraction when no distortion is applied. Among the evaluated settings, the choice of $M = 512$ consistently provides the most favorable trade-off, yielding the strongest overall robustness. In contrast, using a shorter watermark length leads to a noticeable drop in robustness under several attacks, suggesting that the representation capacity is insufficient to provide stable redundancy against distortions. A short watermark length results in too few calibration bits, making watermark extraction more fragile to distortions. On the other hand, further increasing the watermark length beyond this point does not continue to improve performance. An excessively large watermark length reduces the repetition count of each watermark bit, thereby increasing the instability of watermark extraction.

In addition, the synchronization header directly determines the resistance of the watermark to geometric attacks. As shown in Table 13, removing the geometric synchronization causes the accuracy to fall immediately for rotation or flipping.

To further validate the effectiveness of the synchronization header, we conduct a comprehensive ablation study of the synchronization header. As illustrated in Fig. 4, the synchronization header improves the robustness of watermark extraction, especially under strong distortions.

7. Limitations and future work

PDMark currently targets prepared deniability in a single-round post-disclosure setting. Its guarantees do not directly compose across

Table 12
Ablation study on k and M under standard distortion attacks.

Parameter	Clean	Gaussian noise	Gaussian blur	Median filter	JPEG	Brightness	Rotation	Horizontal flip	Scale
$k = 0.5$	0.999	0.958	0.942	0.983	0.962	0.996	0.970	0.994	0.982
$k = 1.0$	1.000	0.977	0.961	0.990	0.978	0.999	0.981	0.998	0.996
$k = 1.15$	1.000	0.988	0.973	0.997	0.989	0.999	0.994	0.997	0.999
$k = 1.25$	0.997	0.928	0.913	0.959	0.930	0.988	0.952	0.997	0.988
$k = 2.0$	0.985	0.836	0.771	0.850	0.838	0.925	0.967	0.998	0.996
$M = 384$	0.998	0.953	0.954	0.984	0.959	0.995	0.980	0.995	0.989
$M = 512$	1.000	0.988	0.973	0.997	0.989	0.999	0.994	0.997	0.999
$M = 768$	1.000	0.927	0.881	0.959	0.929	0.990	0.975	0.993	0.992
$M = 1024$	1.000	0.971	0.919	0.981	0.978	0.998	0.982	0.996	0.995

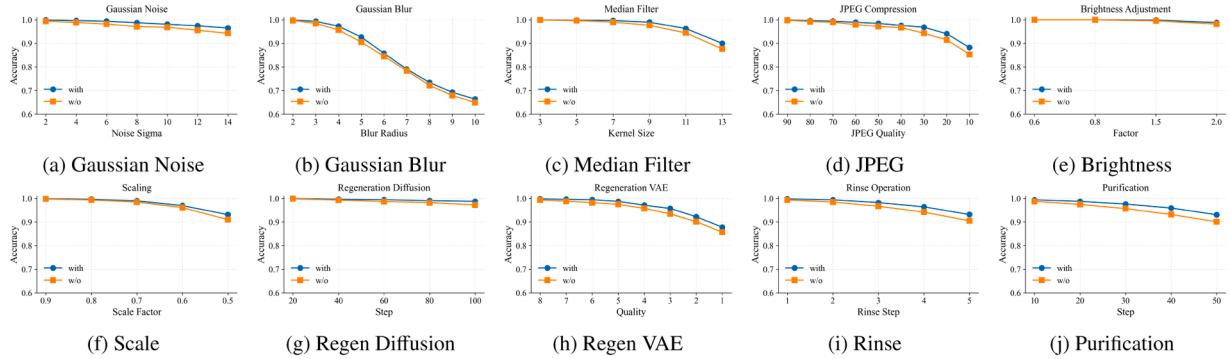


Fig. 4. Ablation study under various distortions.

Table 13
Ablation on geometric synchronization.

	CSP Avg.	Rotation				Flip		
		5°	15°	45°	90°	H	V	HV
with	0.895	0.995	0.989	0.983	0.997	0.997	0.998	0.998
w/o	0.874	0.521	0.504	0.479	0.499	0.503	0.516	0.471

repeated or correlated coercion attempts. In particular, if an adversary forces the user to disclose different keys, messages, or auxiliary records over multiple rounds, the resulting disclosures may become mutually inconsistent or reveal statistical relationships that are not covered by the current analysis. Repeated coercion may also allow the adversary to compare extraction results across images, correlate multiple disclosures associated with the same user or key, and gradually accumulate evidence about the protected branch. The current framework further assumes proper key management and statistically compatible distributions for the real and fake messages, which become more difficult to maintain when disclosures occur repeatedly.

Future work will extend PDMark toward a composable security definition for repeated and adaptive disclosure. We will investigate mechanisms that maintain consistency across multiple coercion rounds, prevent information accumulation from prior disclosures, and limit cross-image linkage. Promising directions include per-image key diversification, key evolution, controlled generation of multiple plausible disclosures, and formal analysis of how much auxiliary evidence can be safely exposed over time. Practical deployment will also require secure meta-data handling and side-channel-resistant implementations.

8. Conclusion

In this paper, we study deniable watermarking for diffusion models and discuss that conventional deterministic verification is insufficient in coercive settings. To address this problem, we proposed PDMark, a train-free latent-based deniable watermarking method that embeds two messages into a watermark representation. Combined with Gaussian-

consistent latent sampling, the proposed design enables standard verification in normal use while supporting deniable disclosure under coercion. Extensive experiments and theoretical analysis demonstrate that PDMark achieves deniability while maintaining high extraction accuracy, competitive visual quality, and strong robustness. Under the stated assumptions and evaluated exact-carrier attack suite, we observe no practically exploitable known-plaintext signal for hidden-branch distinction or hidden plaintext reconstruction.

CRedit authorship contribution statement

Zikai Xu: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization; **Bin Liu:** Writing – review & editing, Funding acquisition; **Weihai Li:** Writing – review & editing, Funding acquisition; **Lijunxian Zhang:** Validation, Investigation; **Nenghai Yu:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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